

TITLE: - KEYBOARD DISABLE

AIM: -

To write a program to disable Keyboard for 5 seconds by masking interrupts and enable it back

ALGORITHM: -

- Initialize data segment.
- Display keyboard disable message.
- Read and save current PIC interrupt mask from port 21h.
- Set bit 1 of the mask to disable IRQ1 (Keyboard) and write to port 21h.
- Read current BIOS timer tick from memory address 0040:006C.
- Add 91 ticks to the current timer value to set a 5-second target time.
- Loop continuously until the memory timer tick reaches the target time.
- Restore the original saved PIC mask to port 21h to re-enable the keyboard.
- Display keyboard enable message.
- Terminate program execution.

PROGRAM:-

TITLE KEYBOARD.ASM -- Disable Keyboard for 5 Seconds Using PIC Masking

```
;/*****  
  
;/*  
  
;/* Copyright (c) 1985-2025, American Megatrends International LLC. *  
  
;/*  
  
;/* All rights reserved. Subject to AMI licensing agreement. *  
  
;/*  
  
;/*****  
  
;<AMI_FHDR_START>  
  
;
```

```
; Header:

; Author: Muthuganesh S

; Description:

; Disable keyboard interrupt (IRQ1) for 5 seconds by masking

; PIC interrupt and enable it back.

;

;<AMI_FHDR_END>
```

```
.model small
```

```
.386
```

```
.stack 100h
```

```
;-----
```

```
; DATA SEGMENT
```

```
;-----
```

```
data segment
```

```
msg1 db 13,10,"Keyboard Disabled for 5 Seconds...",13,10,''
```

```
msg2 db 13,10,"Keyboard Enabled Back.",13,10,''
```

```
oldmask db ?
```

```
data ends
```

```
;-----
```

;-----

; CODE SEGMENT

;-----

code segment

assume cs:code, ds:data

start:

mov ax, data

mov ds, ax

;-----

; Print Disable Message

;-----

mov dx, offset msg1

mov ah, 09h

int 21h

;-----

; Read Current PIC Mask

; PIC Master Mask Port = 21h

; IRQ1 = Keyboard Interrupt

;-----

in al, 21h

mov oldmask, al

;-----

; Mask Keyboard Interrupt (IRQ1)

; Set Bit1 = 1

;-----

or al, 00000010b

out 21h, al

;-----

; Delay for 5 Seconds

; BIOS Timer Tick:

; 18.2 ticks/sec

; 5 sec $\approx$ 91 ticks

; BIOS Data Area Timer = 0040:006C

;-----

mov ax, 0040h

mov es, ax

mov eax, es:[006Ch]

add eax, 91

wait_loop:

cmp es:[006Ch], eax

jnl wait_loop

;-----

; Restore Original PIC Mask

;-----

mov al, oldmask

out 21h, al

;-----

; Print Enable Message

;-----

mov dx, offset msg2

mov ah, 09h

int 21h

;-----

; Exit Program

;-----

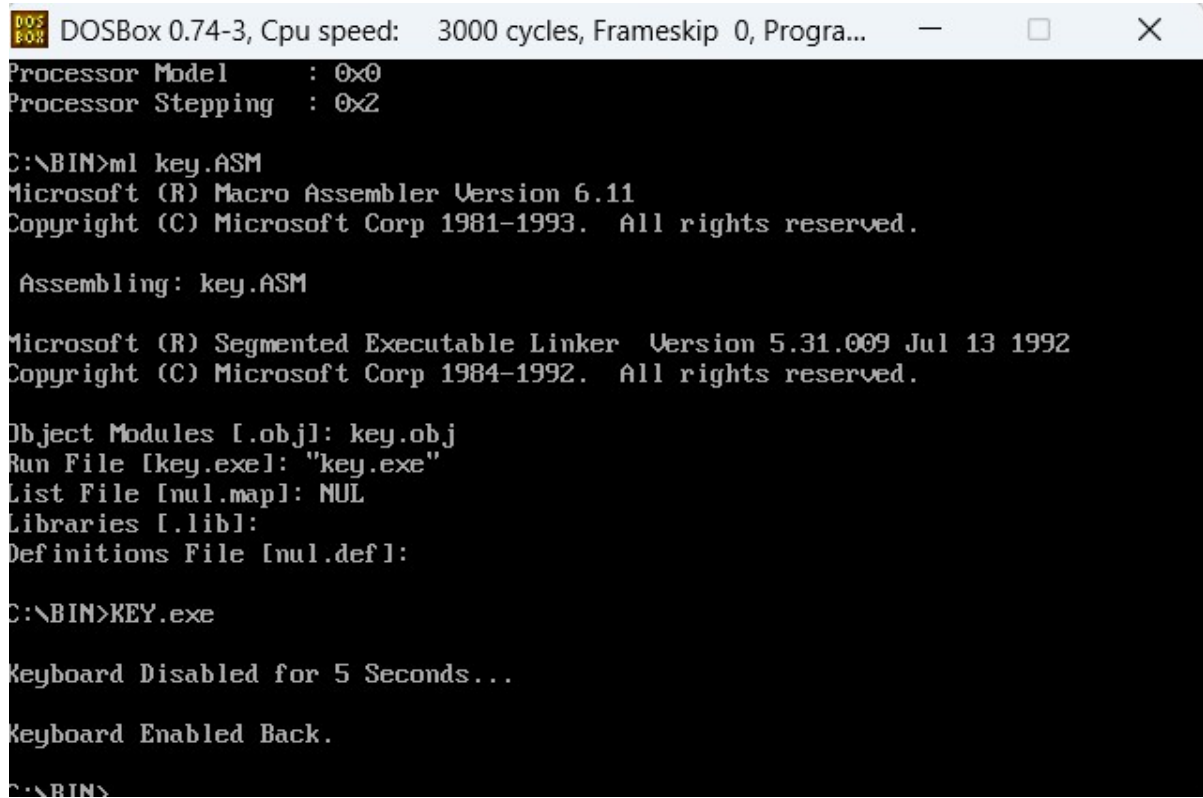
mov ah, 4Ch

int 21h

code ends

end start

OUTPUT: -



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Progra...
Processor Model      : 0x0
Processor Stepping   : 0x2

C:\BIN>ml key.ASM
Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.

Assembling: key.ASM

Microsoft (R) Segmented Executable Linker Version 5.31.009 Jul 13 1992
Copyright (C) Microsoft Corp 1984-1992. All rights reserved.

Object Modules [.obj]: key.obj
Run File [key.exe]: "key.exe"
List File [nul.map]: NUL
Libraries [.lib]:
Definitions File [nul.def]:

C:\BIN>KEY.exe

Keyboard Disabled for 5 Seconds...

Keyboard Enabled Back.

C:\BIN>
```